

AMENDMENT NO. 1 JANUARY 2024
TO
IS 10702 : 2023 HAWAI CHAPPAL
(Third Revision)

(Page 1, clause 2) — Insert the following at the appropriate places:

<i>IS No.</i>	<i>Title</i>
IS 1638 : 1969	Specification for sizes and fittings of footwear (<i>first revision</i>)
IS 3400 (Part 3) : 2021/ISO 4649 : 2017	Methods of test for vulcanized rubbers: Part 3 Abrasion resistance using a rotating cylindrical drum device (<i>third revision</i>)’.

(Page 1, clause 3) — Insert the following new clause and renumber the subsequent clauses:

‘4 CATEGORY AND TYPE

The *hawai chappal* shall be of the following two categories on the basis of their usage:

- a) For kids (standard English sizes as per IS 1638 for infants and children); and
- b) For adults (for all other sizes as per IS 1638).

Further, the *hawai chappal* for adults shall be of two types namely, Type 1 and Type 2.’

(Page 1, clause 4.1.4, second line) — Substitute ‘may’ for ‘shall’.

(Page 2, clause 4.3.1) — Delete.

(Page 2, clause 4.3.2) — Substitute the following for the existing:

‘The thickness of the strap wings for *hawai chappal* may be as agreed to between the purchaser and the manufacturer.’

(Page 2, Fig. 2) — Delete.

(Page 2, clause 4.4) — Substitute the following for the existing clause:

‘4.4 The bottom, strap material and *hawai chappal* shall conform to the requirements as prescribed in Tables 1, 2, and 3 respectively when tested in accordance with the test methods mentioned in col (6) of Tables 1, 2, and 3.’

(Page 2, Fig. 1) — Substitute the following figure for the existing figure:

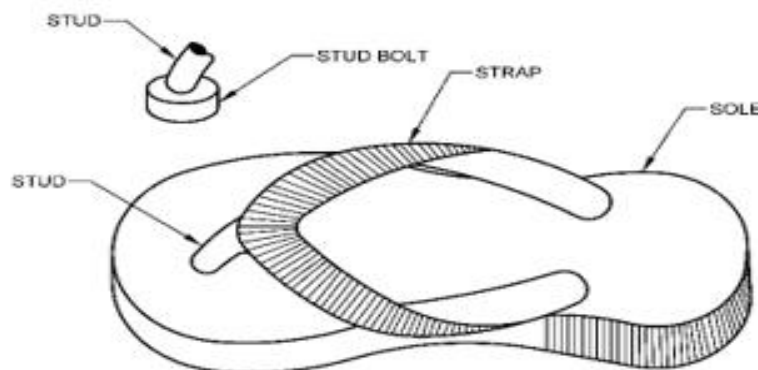


FIG. 1 ILLUSTRATED DIAGRAM OF *HAWAI CHAPPAL* WITH STRAP

Amendment No. 1 to IS 10702 : 2023

(Page 2, Tables 1, 2, and 3) — Substitute the following tables for the existing tables:

Table 1 Physical Requirement of Bottom Material for *Hawai Chappal*

(Clause 4.4)

Sl No.	Characteristic	Requirements			Method of Test, Ref to	
		For Kids ¹	For Adults		Annex	IS
			Type 1	Type 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Compression set at constant stress, percent, <i>Max</i>	65	50	65	D	—
ii)	Split tear strength, kg, <i>Min</i>	—	2.0	—	E	—
iii)	a) Shrinkage at 50 °C ± 1 °C for 2 h for polymeric material ² , percent, <i>Max</i>	—	3.0	5.0	F	—
	b) Shrinkage at 100 °C ± 1 °C for 1 h for rubber, percent, <i>Max</i>	—	3.0	5.0		
iv)	Abrasion resistance at applied force of 5 N, (volume loss), mm ³ , <i>Max</i>	—	700	1 300	—	IS 3400 (Part 3)
v)	Flexing resistance: Maximum cut growth, in mm (test carried out at 25 °C ± 2 °C)	At 15 000 flexes, 6 mm	At 50 000 flexes, 6 mm	At 20 000 flexes, 6 mm	—	IS 3400 (Part 16)
vi)	Hydrolysis resistance: Maximum cut growth (applicable only for polyester based PU bottom)	—	At 50 000 flexes, 6 mm	—	—	IS 15298 (Part 1)
vii)	Chemical requirements ³	The bottom (only in contact with skin) of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as specified under 3.6 of IS 17011	The bottom (only in contact with skin) of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as specified under 3.6 of IS 17011	The bottom (only in contact with skin) of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as specified under 3.6 of IS 17011	—	Table 1 of IS 17011

Amendment No. 1 to IS 10702 : 2023

Sl No.	Characteristic	Requirements			Method of Test, Ref to	
		For Kids ¹	For Adults		Annex	IS
			Type 1	Type 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
viii)	Lead content (as Pb), ppm, <i>Max</i> (applicable for PVC only)	2	2	–	–	12240 (Part 5)
NOTES ¹ In case of infant standard English sizes 1 to 4 as per IS 1638, only chemical requirements as specified as sl no. (vii) shall be tested. ² In case of combination of polymeric and rubber material, the temperature conditions for polymeric material shall be used. ³ In case of Type 2 and kids except infants (as specified in Note 1), this test shall be kept in abeyance till 31st December 2024.						

Table 2 Physical Requirement of Strap Material for *Hawai Chappal*

(Clause 4.4)

SI No.	Characteristic	Requirement			Method of Test, as per	
		For Kids ¹	For Adults		Annex	IS
			Type 1	Type 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Breaking strength, kgf <i>Min</i>				G	–
	a) Strap to strap ²					
	1) Solid strap	15	30	15		
	2) Cellular strap	8	10	10		
	b) Toe portion to strap ²					
	1) Solid strap	10	10	10		
	2) Cellular strap	7	7	7		
	c) Elongation at break ² , percent, <i>Min</i>	–		–	G	–
	1) Solid strap		250			
	2) Cellular strap		100			
[applicable to strap to strap (V shape)]						
ii)	Chemical requirements ³	The strap of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as	The strap of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as	The strap of <i>hawai chappal</i> shall comply with Table 1 of IS 17011 for critical substance Category 1 and Category 2 as	Table 1 of IS 17011	

Amendment No. 1 to IS 10702 : 2023

SI No.	Characteristics	Requirement			Method of Test, as per	
		For Kids ¹	For Adults		Annex	IS
			Type 1	Type 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
		specified under 3.6 of IS 17011	specified under 3.6 of IS 17011	specified under 3.6 of IS 17011		
iii)	Lead content (as Pb), in ppm, <i>Max</i>	2	2	—	IS 12240 (Part 5)	
NOTES						
¹ In case of infants standard English sizes 1 to 4 as per IS 1638, only chemical requirements as specified as sl no. (ii) shall be tested.						
² Solid, density more than 0.90 g/cm ³ ; and Cellular, density less than 0.90g/cm ³ .						
³ In case of Type 2 and kids except infants (as specified in Note 1), this test shall be kept in abeyance till 31 st December 2024.						

Table 3 Attachment Strength and Inter Layer Bond Strength Requirement for *Hawai Chappal*
(Clause 4.4)

Sl No.	Characteristic	Requirement			Method of Test as per	
		For Kids	For Adults		Annex	IS
			Type 1	Type 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Toe post attachment strength, in N, <i>Min</i>	40	80	50	H	—
ii)	Attachment strength of rear strap, in N, <i>Min</i>	40	80	50		
iii)	Inter layer bond strength, in N/mm, <i>Min</i> (applicable only for multilayer pasted stuck-on bottom sole)	0.7	2	1.2	—	IS 15298 (Part 1)

(Page 3, clause 5.1) — Delete.

(Page 3, clause 5.2) — Delete.

(Page 4, clause 6.1.1) — Substitute the following for the existing clause:

‘6.1.1 The *hawai chappal* shall be marked legibly with the following:

- a) Size;
- b) Type; and
- c) Identification of the source of manufacturer or brand name.

NOTE — Kids *hawai chappal* shall be marked as ‘0’ whereas infants *hawai chappal* shall be marked as ‘00’.

(Page 4, clause 6.2) — Substitute the following for the existing clause:

‘6.2 Packing

The *hawai chappal* may be packed as agreed to between the purchaser and the manufacturer. The straps may also be marked with the side of foot, that is, right and left. Each package may contain *chappal* of one size only and

Amendment No. 1 to IS 10702 : 2023

may be marked with the name of the item, size, colour, type, identification of the source of manufacture and batch number.'

(Page 4, clause 7) — Delete.

(Page 6, Annex B) — Delete.

(Page 7, clause C-1.4, second para, third line) — Substitute '27 °C ± 2 °C' for '23 °C ± 2 °C'.

(Page 9, clause C-2.4, third para, second line) — Substitute '27 °C ± 2 °C' for '(25 ± 2) °C'.

(Page 11, clause D-1.4.5, third line) — Substitute '27 °C ± 2 °C' for '20 °C ± 2 °C'.

(Page 12, clause D-1.6) — Substitute the following for the existing :

$$\text{'Percentage of compression set} = \frac{T_0 - T_1}{T_0} \times 100$$

where

T_0 = arithmetic mean of the thickness of test piece in mm, before compression; and

T_1 = arithmetic mean of the thickness of test piece in mm, after compression.'

(Page 12, clause D-2.1.4) — Substitute the following for the existing:

D-2.1.4 Expression of Results

The compression set is the difference between the original thickness of the test piece and that after recovery expressed as a percentage of the initial thickness.

$$\text{Compression set, percent} = \frac{t_0 - t_i}{t_0} \times 100$$

where

t_0 = initial thickness, in mm; and

t_i = final thickness, in mm.'

(Page 14, clause F-5) — Substitute the following for the existing:

'Calculate the shrinkage as follows:

$$\text{Shrinkage, percent} = \frac{L_0 - L_1}{L_0} \times 100$$

where

L_0 = length of test piece, in mm, before heating; and

L_1 = length of test piece, in mm, after heating and cooling at room temperature.'

(Page 15, clause G-4.2, second para, second line) — Substitute '27 °C ± 2 °C and (65 ± 5) percent relative humidity' for '25 °C ± 2 °C and 65 percent ± 5 percent RH'.

(Page 19, clause H-5, second line) — Substitute '27 °C ± 2 °C and (65 ± 5) percent relative humidity' for '(23 ± 2) °C and 50 percent ± 4 percent RH'.

(CHD 19)